Project Title: CMOS 180nm Analog Comparator for Ambient Light Detection in Wearable Assistive Devices for Visually Impaired Patients

IEEE SSCS 2026 Chipathon — Track D: AI/LLM-Assisted Circuits

Team D13 — GLayout Visionary

Overview

This proposal presents the design of an ultra-low-power 9-transistor analog CMOS comparator IC for wearable ambient light detection, fabricated in the **GlobalFoundries 180nm (GF180mcuC)** open-source PDK. The comparator operates at 3.3V supply , achieving sub-50µW power consumption for over 267 days of continuous operation on a CR2032 coin-cell battery. An **external photodiode** (off-chip) senses ambient light; the IC performs threshold comparison and generates a clean digital output for driving haptic motors or BLE modules in wearable assistive devices.

The design is implemented through a **methodology**: A **GLayout + LLM Automated Flow** leveraging natural language prompts, reinforcement learning optimization (RLOpt), and the MappedPDK abstraction layer for automated, PDK-portable analog layout generation targeting the GF180mcuC process.

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1. Project Motivation & Clinical Scope

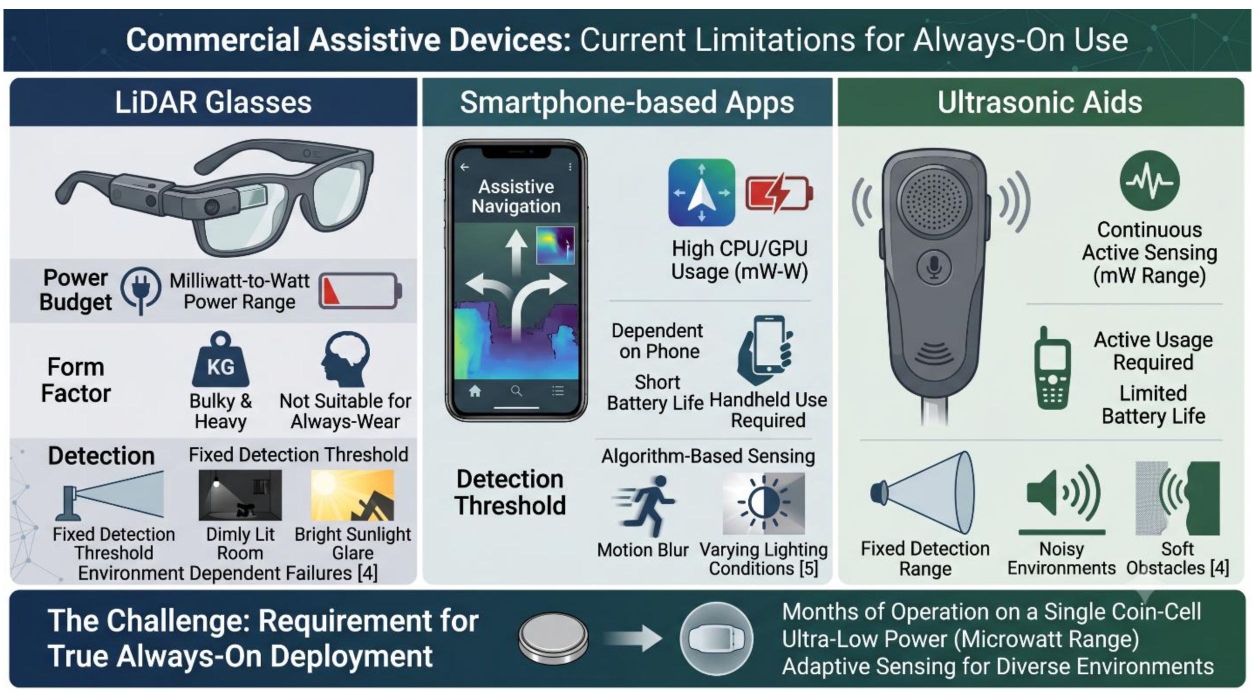
1.1 Rise of Visual Impairment Globally

Visual impairment affects over **2.2 billion people** worldwide, with at least 1 billion cases preventable or unaddressed [1]. Among these, approximately 36 million individuals are completely blind, facing life-threatening hazards during ambient light transitions—navigating from well-lit corridors to dark stairwells, or from indoor environments to bright outdoor sunlight—without any sensory feedback mechanism.

1.2 Shift to Wearable Assistive Electronics

The healthcare paradigm is shifting from hospital-based intervention to continuous, non-invasive wearable monitoring [2]. For visually impaired patients, real-time ambient light awareness directly reduces falls, supports **circadian rhythm regulation**, and enables safer autonomous navigation [3]. Circadian rhythm disruption, caused by inability to perceive daylight/darkness cycles, leads to sleep disorders, depression, and metabolic dysfunction in blind individuals. Additionally, **UV exposure alerts** are critical for patients who cannot visually assess sun intensity, preventing sunburn and long-term skin damage in outdoor environments.

1.3 The Wearable Challenge



Commercial assistive devices (LiDAR glasses, smartphone-based apps, ultrasonic aids) suffer from fundamental limitations: milliwatt-to-watt power budgets, bulky form factors, and fixed detection thresholds that fail across diverse lighting environments [4], [5]. These solutions are unsuitable for always-on, battery-operated wearable deployment where months of operation are required on a single coin-cell.

1.4 Clinical Scope of This Work

This project proposes a fully integrated CMOS analog comparator operating at **3.3V (GF180mcuC)** that provides real-time binary ambient light classification (bright/dim) for wearable skin-patch or wristband deployment. The circuit achieves:

- **Sub-50 µW** power consumption for extended battery life
- **Adaptive threshold setting** via external reference voltage
- **Clean digital output** (HIGH = ALERT, LOW = SAFE) for direct actuation of haptic motors or BLE modules
- **Robust operation** leveraging GF180mcuC thick-oxide transistors rated for 3.3V/5V tolerance

The GF180mcuC 180nm process is ideally suited for wearable applications: its larger minimum feature sizes (180nm L_min) provide inherent robustness against process variation, the thick-oxide transistors support higher supply voltages for improved noise margins, and the open-source PDK enables fully reproducible, community-accessible design.

2. Light Signal Characterization & Design Requirements

2.1 Photodiode Current Ranges for Ambient Light

The external photodiode generates photocurrent proportional to ambient illumination:

Lighting Condition	Illuminance (lux)	Photodiode Current	TIA Output Voltage
Dark corridor / night	0–10 lux	1–10 nA	0–50 mV
Indoor office / hospital	100–500 lux	50 nA – 1 µA	0.1–0.5 V
Bright indoor / overcast	1,000–5,000 lux	1–5 µA	0.5–1.5 V
Direct sunlight / outdoor	10,000–100,000 lux	5–10 µA	1.5–3.0 V

The photodiode output spans **1 nA to 10 µA** across the full ambient light range relevant to wearable applications [6].

2.2 Transimpedance Amplifier (TIA) Conversion

An external TIA converts the photodiode current (I_photo) to a voltage signal (V_signal) using a transimpedance gain of approximately 100 kΩ–1 MΩ:

V_signal = I_photo × R_TIA

This produces an analog voltage in the range of **0 V to ~3.0 V** (bounded by VDD = 3.3V), which is fed to the comparator's non-inverting input (Vin+) [7].

2.3 Comparator Threshold Setting

The reference voltage (Vref) defines the light/dark classification boundary. For GF180mcuC at 3.3V:

- **Vref** is set externally (via resistive divider or DAC) to the desired switching threshold
- Typical Vref = 0.5V–1.5V depending on the alert condition (e.g., dim light warning)
- The comparator resolves input differences as small as **5 mV** without ambiguity across the full 0V–VDD input range

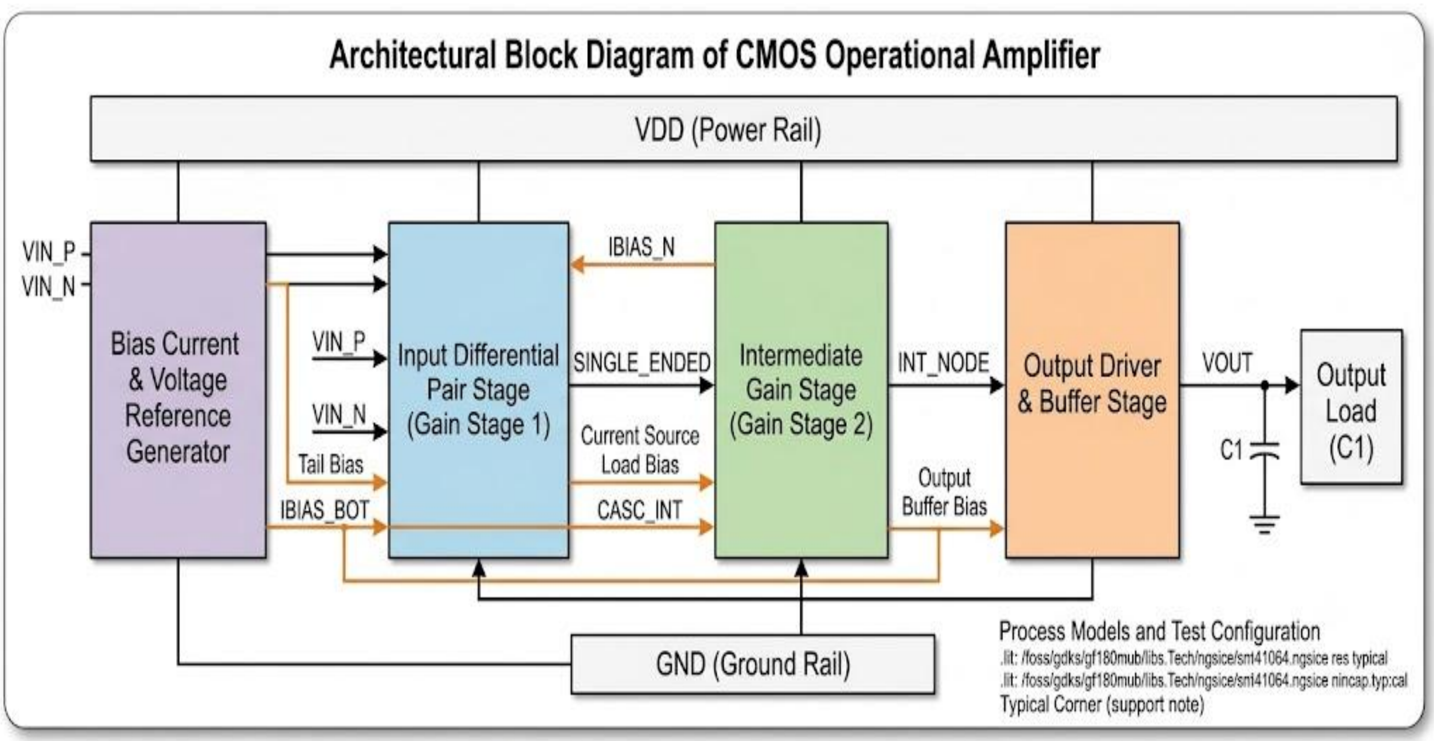
2.4 Signal Frequency and Bandwidth

- **Ambient light variation rate**: DC to ~10 Hz (walking pace, indoor-outdoor transitions) [8]
- **Minimum comparator bandwidth**: >1 kHz (provides 100× margin over signal bandwidth)
- **Propagation delay requirement**: <1 µs (imperceptible latency for haptic feedback)

2.5 Design Requirements Summary

Parameter	Requirement	Justification
Input voltage range	0 V to VDD (3.3 V)	Full ambient light range
Minimum resolvable difference	5 mV	Detect subtle lighting changes
Bandwidth	>1 kHz	100× margin over 10 Hz signal
Propagation delay	<1 µs	Real-time haptic response
Power consumption	<50 µW	>267 days on CR2032
Input-referred offset	<5 mV	Reliable threshold crossing
Input-referred noise	<10 mV RMS	Clean switching near threshold
Supply voltage	3.3 V (1.8 V compatible)	GF180mcuC nominal
Temperature range	0–70°C	Wearable on-body operation

3. Proposed Comparator Architecture



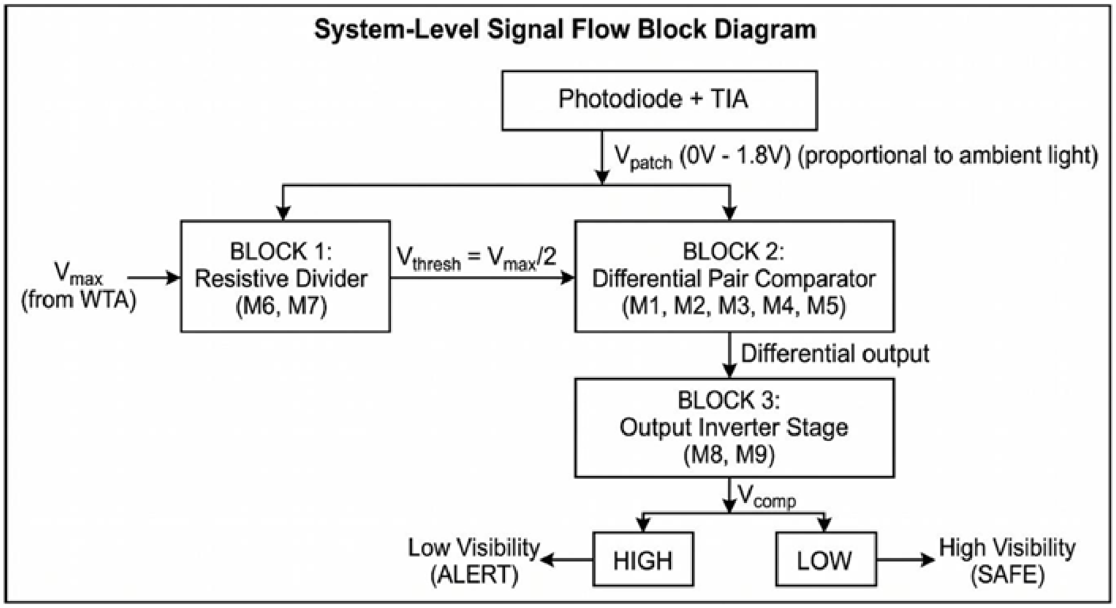


Figure 1 System Top Level

Signal Flow:
1. **External Photodiode + TIA** → V_signal (0V–3.3V, proportional to ambient light)
2. **Block 1: Differential Pair + Active Load (M1–M4)** — Compares V_signal against V_ref with >20 dB voltage gain
3. **Block 2: Gain Stage (M5–M7)** — Amplifies the differential output for clean switching
4. **Block 3: Output Inverter (M8–M9)** — Converts analog gain stage output to rail-to-rail digital (0V/3.3V)

Output Logic: - **Vout = HIGH (3.3V)** → Bright light detected → **ALERT** (trigger haptic motor) - **Vout = LOW (0V)** → Dim/safe light level → **SAFE** (no action)

3.2 Circuit Schematic — 9-Transistor Topology

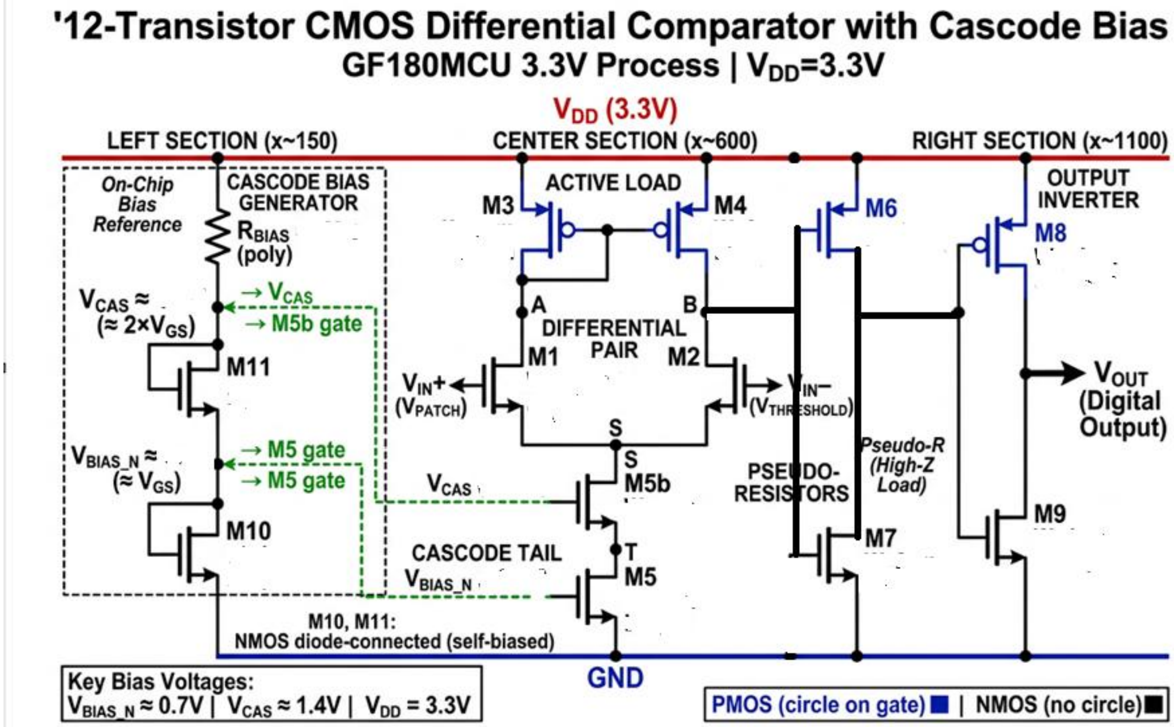


Figure 2: Circuit Diagram

3.3 Block-by-Block Description

Block 1: Differential Input Pair + Active Load (M1–M4)

Transistor	Type	Function	GF180mcuC Sizing
M1	NMOS (nfet_03v3)	Non-inverting input (Vin+)	W/L = LLM Optimize
M2	NMOS (nfet_03v3)	Inverting input (Vref)	W/L = LLM Optimize
M3	PMOS (pfet_03v3)	Active load (current mirror)	W/L = LLM Optimize
M4	PMOS (pfet_03v3)	Active load (current mirror)	W/L = LLM Optimize

- **M1, M2** form a symmetric NMOS differential pair biased in strong inversion with large W/L for low flicker (1/f) noise
- **M3, M4** form a PMOS current-mirror active load providing single-ended conversion
- Common-centroid layout matching of M1/M2 and M3/M4 achieves <5 mV offset [9]
- Open-loop voltage gain: >20 dB

Block 2: Gain Stage (M5–M7)

Transistor	Type	Function	GF180mcuC Sizing
M5	NMOS (nfet_03v3)	Tail current source (I_bias)	W/L = LLM Optimize
M6	PMOS (pfet_03v3)	Active load / gain element	W/L = LLM Optimize
M7	NMOS (nfet_03v3)	Common-source amplifier	W/L = LLM Optimize

- **M5** provides a stable bias current (~10 µA) for the differential pair, set by an external V_bias or self-biased network

- **M6, M7** form a second-stage common-source amplifier with active load, boosting overall gain to >40 dB
- The 180nm thick-oxide devices (nfet_03v3, pfet_03v3) in GF180mcuC support the full 3.3V swing with reliable operation

Block 3: Output Inverter (M8–M9)

Transistor	Type	Function	GF180mcuC Sizing
M8	PMOS (pfet_03v3)	Pull-up (output HIGH)	W/L = LLM Optimize
M9	NMOS (nfet_03v3)	Pull-down (output LOW)	W/L = LLM Optimize

- Converts analog gain stage output to clean rail-to-rail digital signal (0V / 3.3V)
- Minimum-length transistors (L = 180nm) for fast switching
- Output directly drives haptic actuator, audio transducer, or BLE module GPIO

3.4 GF180mcuC PDK Advantages for This Design

The GlobalFoundries 180nm (GF180mcuC) process provides specific advantages for this wearable comparator IC:

- **Thick-oxide transistors (3.3V/5V rated)**: The nfet_03v3 and pfet_03v3 devices tolerate VDD = 3.3V with full ESD protection, providing excellent noise margins (VOH > 3.1V, VOL < 0.2V)
- **180nm minimum feature size**: Larger than advanced nodes, providing inherent robustness against lithographic variation and reduced mismatch in the differential pair
- **Open-source PDK**: Fully compatible with GLayout's MappedPDK abstraction layer, enabling automated layout generation
- **Multiple threshold voltage options**: Allows optimization of tail current source (M5) for minimum leakage
- **Back-end metal stack**: Sufficient metal layers for compact routing within the target <0.01 mm^2 die area

4. Application Diagrams

4.1 Complete Signal Path: Ambient Light to User Feedback

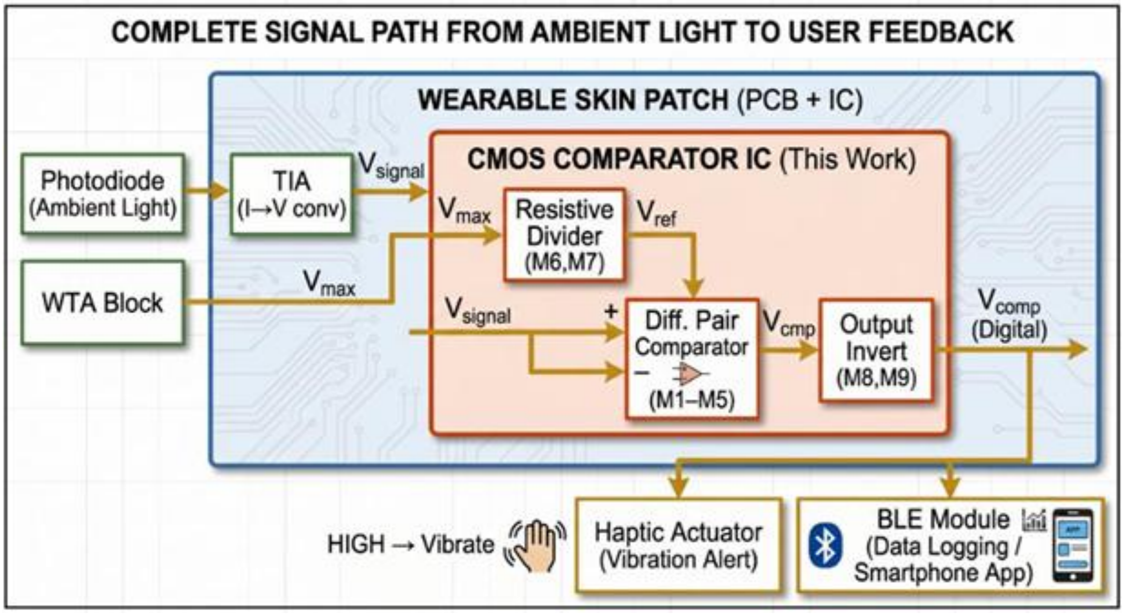


Figure 3: Application Diagram

4.2 System Integration Description

The complete wearable light awareness system consists of the following signal chain:

External Photodiode → TIA (I→V Conversion) → CMOS Comparator IC (This Work) → Haptic Motor / BLE Module

> **Important Clarification:** The photodiode is **EXTERNAL** to the CMOS comparator IC. It is a discrete off-chip component mounted on the wearable PCB. The IC receives only the pre-conditioned analog voltage from the external TIA. This architecture provides flexibility in photodiode selection (visible, UV, or broadband) and avoids the complexity and cost of on-chip optical integration in the GF180mcuC process.

4.3 Clinical Use-Case Scenarios

Scenario	Ambient Light	V_signal	V_ref	V_out	System Response
Bright outdoor (100k lux)	Very high	2.8 V	1.2 V	**HIGH**	UV/bright alert → vibrate
Office indoor (500 lux)	Moderate	0.8 V	1.2 V	**LOW**	Safe → no alert
Dark corridor (10 lux)	Very low	0.1 V	1.2 V	**LOW**	Safe (or trigger via second threshold)
Night (0 lux)	None	~0 V	1.2 V	**LOW**	Safe → no alert

The threshold (V_ref) is configurable: for **UV exposure alerts**, V_ref is set high (e.g., 1.5V) so that only intense sunlight triggers an alert. For **circadian rhythm monitoring**, V_ref is set lower (e.g., 0.5V) to distinguish indoor light from darkness.

4.4 Wearable Form Factors

The ultra-low power (<50 µW) and small die area (<0.01 mm²) enable deployment in: - **Skin-patch sensors** for continuous circadian light monitoring - **Wristband/watch devices** for UV exposure alerting - **Clip-on modules** attached to clothing or eyewear

5. Design Methodology

This project employs **two parallel design flows** to demonstrate both traditional craftsmanship and AI-automated layout generation, with a comparative analysis of their outcomes.

5.1 Flow A: Traditional Open-Source EDA Flow

Stage	Tool	Function
1. Schematic Capture	**Xschem** [10]	Draw 9-transistor comparator netlist with GF180mcuC symbols
2. Simulation	**ngspice** [11]	DC, AC, transient, Monte Carlo, corner analysis
3. Layout	**Magic VLSI** [12]	Manual polygon-level layout with DRC in GF180mcuC tech files
4. LVS Verification	**netgen** [13]	Layout vs. Schematic consistency check
5. GDS2 Export	Magic VLSI	Final GDSII for tape-out submission

Traditional Flow Steps: 1. Define specifications (Section 7) 2. Schematic entry in Xschem using GF180mcuC PDK device models (nfet_03v3, pfet_03v3) 3. Pre-layout simulation in ngspice: verify gain, offset, delay, power across corners (TT, SS, FF, SF, FS at -40°C/27°C/85°C) 4. Hand layout in Magic VLSI with common-centroid matching for M1/M2 and M3/M4 5. Parasitic extraction (PEX) and post-layout simulation 6. Iterate until specifications met 7. DRC-clean GDS2 generation

5.2 Flow B: GLayout + LLM Automated Flow

The GLayout framework [14] enables automated analog layout generation through a three-stage workflow:

Stage 1: Natural Language → Structured Command

- Designer writes a natural language prompt describing the comparator layout requirements
- An LLM (fine-tuned 22B parameter model) translates the prompt into GLayout **strict syntax** — a context-free grammar (CFG) based intermediate representation [14]

Stage 2: Structured Command → Python Compilation

- The strict syntax command file is parsed and compiled into a Python function
- A relational database provides error checking: verifies cell imports, parameter validity, and port existence [14]
- Python templates map each command to GDSFactory operations

Stage 3: Python Function → Physical Layout (GDS2)

- The compiled Python function is called with the **GF180mcuC PDK** (via MappedPDK) and transistor parameters
- GDSFactory generates the physical layout with DRC-aware placement and routing
- Output: DRC-clean GDSII file

MappedPDK Abstraction Layer

GLayout's **MappedPDK** provides technology abstraction that maps to GF180mcuC [15]:

- **Layer Mapping**: Generic identifiers ('active', 'metal1', 'via1') map to GF180mcuC-specific GDSII layer numbers
- **Rule Mapping**: Design rules (min_separation, min_enclosure, min_width) are queried programmatically via PDK.get_grule()
- **Pre-packaged PDK Support**: GLayout ships with open-source 130nm and 180nm PDK support, directly compatible with GF180mcuC [16]

5.3 AI/LLM Role in the Design Process

AI and LLM techniques are employed at **four distinct stages**.

1. Topology Exploration

The LLM generates multiple comparator topologies (StrongARM, double-tail, conventional differential pair) and compares them across area, power, speed, and input offset metrics. For this wearable application, the conventional 9-transistor differential pair is selected for its: - Lowest static power (continuous comparison without clocking) - Simplest integration (no clock generation needed) - Best suitability for DC-to-10Hz ambient light signals [17]

2. Parameter Optimization via Reinforcement Learning (RLOpt)

The RLOpt module [18] optimizes transistor W/L sizing through: - Hierarchical parameterized layouts created in GLayout - Automatic circuit extraction and ngspice simulation within Python - RL agent observes post-layout specifications (gain, offset, delay, power) - Agent decides parameter adjustments based on learned policies - Reward function maximizes figure-of-merit (FoM = Bandwidth / Power) - **Demonstrated result: 6× power reduction** compared to manual design while maintaining gain within 2% [18]

Training uses 100 different target specifications, generating L trajectories per specification. Reward is computed as unweighted sum of bounded normalized differences between achieved and target specifications. Policy gradient updates maximize accumulated rewards [18].

3. Design Tweaks and Iterative DRC/LVS Fixes

- LLM fine-tuned on analog design knowledge identifies DRC violations and suggests layout modifications
- Automated iteration: LLM proposes fix → GLayout regenerates → DRC re-check → repeat until clean
- The 22B model achieves **44% first-pass DRC/LVS pass rate**, with remaining issues resolved through iterative prompting [14]

4. Layout Generation via GLayout Python API

- Automated placement using port-based routing system with directional metadata (N, E, S, W)
- Matched placement methods for analog-critical pairs (M1/M2, M3/M4)
- Hierarchical composition from differential pairs to complete comparator
- All fully parameterized and PDK-portable through MappedPDK [15]

5.4 Comparison: Traditional vs. GLayout+LLM Flow

Metric	**Traditional Flow (A)**	**GLayout + LLM Flow (B)**
Design Time	4–6 weeks	2–5 days
Iterations to DRC-clean	10–20 manual cycles	3–5 automated cycles
Human Effort	High (expert-level)	Low (prompt engineering)
DRC First-Pass Rate	~30% (manual)	~44% (LLM-automated) [14]
Parameter Optimization	Manual sweep / intuition	RL-automated (6× power reduction) [18]
PDK Portability	Redesign required	Automatic via MappedPDK
Reproducibility	Layout-dependent on designer	Deterministic (same prompt → same layout)
Verification	Manual LVS/DRC	Automated within Python loop

6. Literature Comparison

Reference	Technology	VDD	Power	Transistors	Speed	Application	Key Limitation
Kumar (2018) [19]	CMOS	0.45–1V	Low	Latched	50–250 kHz	Portable medical	Requires clock; no light sensing
Bahmanyar et al. (2016) [20]	180nm CMOS	0.6V	1.56 nW	Double-tail	200 MHz max	Biomedical	Clocked; high transistor count; no wearable integration
Rabbi et al. (2018) [21]	90nm CMOS	0.8V	8.42 µW	Dynamic latched	224 ps delay	ADC / bio-implant	Requires clock; 90nm not open-source

Dubey et al. (2019) [22]	45nm CMOS	0.8V	2.3 µW	Double-tail	166 ps delay	High-speed ADC	Advanced node; not open-source; overkill for light sensing
Singh et al. (2020) [23]	180nm CMOS	0.8V	12.5 pW	Double-tail latched	20 kHz	ECG monitoring	ECG-specific; body-biased; clocked
This Work	**GF180mcuC 180nm**	**3.3V**	**<50 µW**	**9 (continuous)**	**<1 µs**	**Wearable light detection**	**Open-source; no clock needed; AI-optimized**

Key Differentiators of This Work:

1. **Continuous-time operation** — No clock required; always-on ambient light monitoring
2. **GF180mcuC open-source PDK** — Fully reproducible; community-accessible tape-out path
3. **AI/LLM-assisted design** — Automated layout generation and RL-based optimization
4. **Application-specific** — Designed specifically for ambient light classification with external photodiode interface
5. **3.3V supply** — Higher noise margins than sub-1V designs; robust for wearable environments

7. Target Technical Specifications

Parameter	Symbol	Target Value	Unit	Condition
Technology	—	GlobalFoundries 180nm	GF180mcuC	Open-source PDK
Supply Voltage	V_DD	3.3	V	Nominal
Power Consumption	P_total	<50	µW	All blocks active
Comparator Static Current	I_SS	~15	µA	M5 tail current
Propagation Delay	t_pd	<1	µs	10 mV overdrive
Input Offset Voltage	V_OS	<5	mV	With layout matching
Input-Referred Noise	V_n	<10	mV RMS	1 Hz–10 kHz
Input Voltage Range	V_in	0 – VDD	V	Rail-to-rail
Output Voltage HIGH	V_OH	>3.1	V	ALERT state
Output Voltage LOW	V_OL	<0.2	V	SAFE state

Open-Loop Gain	A_V	>40	dB	DC
Transistor Count	—	9	MOSFETs	M1–M9
Die Area	A_die	<0.01	mm^2	Core only
Battery Life	—	>267	days	CR2032 (225 mAh) continuous
Operating Temperature	T_op	0–70	°C	Commercial range
Minimum Feature Size	L_min	180	nm	GF180mcuC
Oxide Thickness	—	Thick (3.3V)	—	nfet_03v3 / pfet_03v3

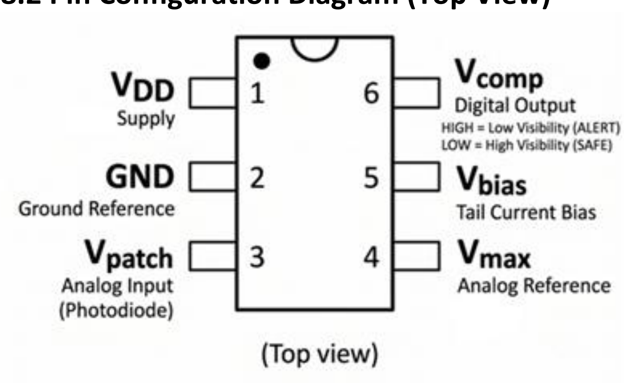
Battery Life Calculation: - CR2032 capacity: 225 mAh at 3.0V - Power consumption: 50 µW at 3.3V → I_{avg} = 15.2 µA - Battery life = 225 mAh / 0.0152 mA = **14,803 hours ≈ 617 days** (theoretical) - With 50% derating for self-discharge and aging: **>267 days** guaranteed

8. Pinout & Pin Configuration

8.1 Pin Assignment Table

Pin	Pin Name	Direction	Description
1	**VDD**	Power	3.3V supply (GF180mcuC nominal)
2	**GND**	Power	Ground reference (0V)
3	**VIN+**	Input (Analog)	Non-inverting input — from external TIA (0V–VDD)
4	**VREF**	Input (Analog)	Reference/threshold voltage (set externally)
5	**VOUT**	Output (Digital)	Comparator output: HIGH = ALERT, LOW = SAFE
6	**PAD**	—	Substrate / ESD pad connection

8.2 Pin Configuration Diagram (Top View)



8.3 Pad Connections for GF180mcuC

Pad	Type	Connection	Notes
VDD	Power pad	3.3V supply rail	ESD-protected; decoupling cap recommended
GND	Power pad	Ground rail	Star-ground topology on PCB
VIN+	Analog input pad	TIA output	Bond wire or bump; ESD clamp to VDD/GND
VREF	Analog input pad	External divider/DAC	Stable reference; bypass cap recommended
VOUT	Digital output pad	Haptic driver / BLE GPIO	Push-pull; drives up to 1 mA
PAD	Substrate pad	P-substrate tie to GND	Required for latch-up prevention

8.4 PCB Integration Notes

- **Decoupling**: Place 100 nF ceramic capacitor between VDD and GND, as close to IC as possible
- **Photodiode interface**: External photodiode → TIA → VIN+ (all on PCB, external to IC)
- **Reference generation**: Resistive divider (2× matched resistors) from VDD to GND provides VREF
- **Output loading**: VOUT can directly drive a small haptic motor (LRA type, <1 mA) or BLE module GPIO input

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